SOUTHEAST UNIVERSITY, BANGLADESH

Course Name: Artificial Intelligence

Course Code: CSE 365.7

Section: 07

Project Title: Comparative Analysis of Machine Learning Models for Diabetes Risk Prediction Using CDC BRFSS 2015 Health Survey Data

Submitted By:

Mymuna Jarin Rifti (2024000000270)

Umme Sanjina Jarin (2023200000796)

Submitted To:

[KMU] Khandaker Mohammad Mohi Uddin

Comparative Analysis of Machine Learning Models for Diabetes Risk Prediction Using CDC BRFSS 2015 Health Survey Data

Mymuna Jarin Rifti

Student ID: 2024000000270 | Department of Computer Science, Southeast University, Bangladesh

2024000000270@seu.edu.bd **Umme Sanjina Jarin**

Student ID: 2023200000796 | Department of Computer Science, Southeast University, Bangladesh

2023200000796@seu.edu.bd March 2026

Abstract

Background: Diabetes mellitus represents one of the most rapidly escalating non-communicable diseases globally, with its undetected progression causing irreversible complications including cardiovascular disease, neuropathy, retinopathy, and chronic kidney disease. Conventional screening methods such as fasting plasma glucose tests are laboratory-dependent and financially prohibitive at scale, rendering them inaccessible in low- and middle-income countries where over 80% of cases remain undiagnosed. **Problem:** A critical gap exists in the literature concerning systematic, cross-formulation comparisons of classical machine learning approaches on large-scale self-reported health survey data while simultaneously addressing class imbalance — a persistent challenge that degrades minority-class recall in deployed systems. **Methodology:** This study addresses these limitations by systematically evaluating Logistic Regression, Random Forest, and Gradient Boosting across three clinically motivated formulations derived from the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset (253,680 instances; 21 features): a threeclass severity formulation, balanced binary classification, and imbalanced binary prediction. Class-weight balancing was employed as a pragmatic imbalance-mitigation strategy. **Results:** Gradient Boosting achieved the highest macro F1-score of 0.5908 on imbalanced binary data, while Logistic Regression outperformed on the balanced formulation with F1 = 0.7483. Feature importance analyses consistently identified BMI, age category, general health status, and hypertension as dominant predictors, aligning with established epidemiological evidence. **Conclusion:** An interactive clinical decision support system was developed and deployed to operationalize the validated model for real-time, laboratory-free diabetes risk assessment. These findings establish actionable baselines for BRFSS-based screening and demonstrate that class-weight balancing offers competitive performance versus complex resampling, advancing scalable and interpretable clinical deployment.

Keywords: diabetes risk prediction; machine learning; BRFSS 2015; class imbalance; Gradient Boosting; Logistic Regression; Random Forest; clinical decision support

1. Introduction

Type 2 diabetes mellitus (T2DM) constitutes a global public health emergency of unprecedented proportions. The International Diabetes Federation projects that T2DM prevalence will affect approximately 12.2% of the world's adult population by 2045, up from 10.5% as of 2021 [1]. The disease inflicts an estimated USD 966 billion annually in healthcare expenditure, with disproportionate burden borne by low- and middle-income countries (LMICs) where laboratory infrastructure for early detection is severely constrained [1]. Complications arising from delayed or missed diagnosis — including cardiovascular disease, peripheral neuropathy, diabetic retinopathy, and end-stage renal disease — not only diminish quality of life but impose catastrophic costs on healthcare systems already operating beyond capacity [2,3].

Problem Motivation: The traditional paradigm for diabetes screening relies on fasting plasma glucose measurement and oral glucose tolerance tests — both of which require clinical infrastructure unavailable across large segments of the global population [4]. Questionnaire-based surveys such as the CDC BRFSS, which collect self-reported health indicators at population scale, represent an underutilised resource for noninvasive screening. Despite the availability of the BRFSS 2015 dataset comprising over 250,000 respondents with 21 clinically grounded features, systematic benchmarking of machine learning approaches across multiple dataset formulations — including multi-class severity stratification and binary risk classification under both balanced and natural class distributions — remains largely absent from the literature [5,6].

Research Gap: Prior studies have demonstrated the applicability of individual machine learning algorithms to diabetes datasets [7,8]; however, critical gaps persist. First, most evaluations employ a single dataset formulation, obscuring the influence of data engineering decisions on model behaviour. Second, class imbalance — inherent in the BRFSS dataset at an 86:14 ratio — is frequently unaddressed or treated with computationally intensive oversampling methods such as SMOTE [9], which may introduce synthetic data artefacts and fail in resource-constrained deployment environments. Third, deployment-ready implementations accompanied by clinically interpretable feature analyses are rarely provided alongside benchmarking results, creating a persistent research-to-practice gap [10].

Our Approach: This study proposes a systematic evaluation framework applied to three clinically motivated BRFSS 2015 formulations. We employ class-weight balancing as a pragmatic, computationally efficient imbalance-mitigation strategy and conduct comprehensive feature importance analysis to validate alignment with established diabetes epidemiology. An interactive clinical decision support system is subsequently deployed to demonstrate real-world applicability of the best-performing validated model.

1.1 Key Contributions

- Systematic benchmarking of Logistic Regression, Random Forest, and Gradient Boosting across three BRFSS 2015 formulations (3-class, balanced binary, imbalanced binary) with consistent preprocessing pipelines and evaluation protocols.
- Empirical validation of class-weight balancing as a competitive and deployment-friendly alternative to SMOTE-based oversampling for clinical imbalance handling on large-scale survey data.

- Feature importance analysis aligned with established diabetes epidemiology (BMI, age, hypertension), enhancing interpretability and clinical trust in the proposed screening models.
- Development and deployment of an interactive clinical decision support system enabling real-time, laboratory-free diabetes risk stratification at population scale.

The subsequent sections are structured as follows: Section 2 reviews representative prior work on machine learning-based diabetes prediction. Section 3 presents the methodology including dataset descriptions, preprocessing pipeline, model architectures, and evaluation framework. Section 4 reports experimental results, confusion matrix analyses, ROC-AUC scores, and comparative performance benchmarks. Section 5 discusses clinical implications and limitations, and Section 6 concludes with future research directions.

2. Related Works

The application of machine learning to diabetes prediction has attracted substantial scholarly attention across diverse datasets and clinical contexts. This section reviews fifteen representative studies, examining their methodological approaches, datasets, best-reported results, and recognised limitations.

Chowdhury et al. [2] investigated twelve machine learning algorithms applied to the BRFSS 2021 dataset (n > 400,000) to predict diabetes risk. Using XGBoost, Random Forest, Logistic Regression, and SVM among others, XGBoost achieved 82% overall accuracy; however, diabetic class recall remained critically low at 32% due to the inherent 86:14 class imbalance. The study demonstrated the inadequacy of accuracy as a sole evaluation metric for imbalanced medical datasets but did not apply systematic imbalance correction, limiting clinical deployment utility.

Yu et al. [3] employed Gradient Boosting on a BRFSS-Tennessee 2023 sub-sample (n ≈ 8,000) to analyse diabetes prediction, reporting 81.5% accuracy but an F1-score of only 0.37 for the diabetes class, illustrating the accuracy-F1 disconnect prevalent in imbalanced settings. The authors advocated macro F1 as the primary evaluation metric for imbalanced health survey data; however, the small regional sample limits generalisability and no deployment interface was provided.

Al-Kubaisi et al. [4] analysed the exact BRFSS 2015 dataset used in this study using Logistic Regression, Random Forest, and ensemble voting, achieving 78% sensitivity for the diabetes class through probability threshold tuning at 0.55. The study validated top predictive features consistent with those identified herein; however, no multi-formulation comparison or interactive deployment was provided.

Wu et al. [5] proposed a GA-XGBoost stacking ensemble with SMOTEENN oversampling on BRFSS 2015 data, achieving AUC = 0.90 and minority-class F1 = 0.62. While demonstrating the value of genetic algorithm-based feature selection combined with advanced ensembles, the high computational complexity limits deployment in resource-constrained clinical environments.

Geasela et al. [6] evaluated Random Forest with minimal feature selection on imbalanced BRFSS data, reporting 86% accuracy but only 23% diabetic class recall. The study confirmed that feature reduction can maintain accuracy, but critically low minority-class performance due to absent class balancing remains a key limitation.

Kavakiotis et al. [7] conducted a systematic meta-analysis of 85 diabetes prediction studies, identifying SVM and neural networks as the most frequently reported algorithms with average accuracy of approximately 78%. This seminal review established cross-study benchmarks; however, its reliance on older, smaller datasets such as the PIMA Indians Diabetes Dataset limits applicability to modern large-scale population surveys.

Zou et al. [8] applied decision trees and feedforward neural networks to electronic health records, reporting 84.1% neural network accuracy and identifying glucose and BMI as top features via decision tree analysis. While demonstrating that interpretable tree models can rival neural networks clinically, EHR-specific feature sets are not generalisable to self-reported survey data such as BRFSS.

Maniruzzaman et al. [9] investigated Gaussian process-based probabilistic classifiers on the PIMA Indians Diabetes Dataset (n = 768), achieving 82.35% accuracy with Linear Discriminant Analysis. Although introducing probabilistic classifiers as interpretable tools for small medical datasets, the PIMA dataset's small size and demographic homogeneity (female, Pima Indian heritage) preclude transfer to populationlevel survey analysis.

Islam et al. [10] combined Random Forest and XGBoost ensemble methods for diabetes risk stratification in a South Asian clinical cohort (n = 520), achieving F1 = 0.79 for the diabetic class on a balanced dataset. As one of few studies addressing South Asian demographics — directly relevant to the current authors' institutional context — this work nonetheless suffers from extremely limited dataset size and absence of public reproducibility materials.

Sisodia and Sisodia [11] evaluated Naive Bayes, Decision Trees, and SVM on the PIMA dataset, with SVM achieving 76.3% accuracy, establishing widely cited classical baselines. The dataset's constraint to female Pima Indian patients and its size of 768 instances severely limits generalisability beyond the benchmark setting.

Acharya et al. [12] proposed a hybrid Random Forest model with SMOTE augmentation on a UCI diabetes dataset, improving diabetic recall by 17% versus no augmentation and achieving AUC = 0.88. However, synthetic data artefacts from SMOTE can introduce noise, and the approach was not evaluated on BRFSSscale datasets.

Mienye and Jere [13] developed an optimised ensemble learning approach for heart disease prediction integrating SHAP for interpretability, achieving 98.4% accuracy on the Cleveland dataset with XGBoost optimised via Bayesian hyperparameter tuning. While primarily focused on cardiovascular disease, this work demonstrates the combined power of ensemble methods with explainability techniques applicable to chronic disease screening.

Talukder et al. [14] proposed the HXAI-ML framework combining hybrid data balancing (Random Oversampling + Tomek Links) with Extra Trees Classifier and XAI techniques (SHAP, LIME, PIA) for cardiovascular disease prediction, achieving 99.24% accuracy on the Framingham dataset. This seminal work directly informs the present study's methodology, particularly its emphasis on hybrid balancing strategies and the integration of explainability tools into clinical prediction frameworks.

Omotehinwa et al. [15] employed Bayesian optimisation to enhance LightGBM for coronary heart disease detection on the Framingham dataset, achieving 98.82% accuracy and AUC = 0.99. The demonstrated effectiveness of advanced hyperparameter tuning with gradient boosting methods provides a strong methodological precedent for the present study's use of Gradient Boosting for imbalanced health classification.

Collectively, these studies establish that while machine learning can achieve high overall accuracy on diabetes and related chronic disease datasets, macro F1-score and minority-class recall are the critical metrics for clinical relevance. Class imbalance remains an unresolved challenge across BRFSS-based studies, and cross-formulation benchmarking with deployment-ready implementations is largely absent from the existing literature. The present work directly addresses these gaps.

3. Methodology

The proposed framework for diabetes risk prediction integrates systematic data formulation, preprocessing, class imbalance mitigation, ensemble and linear classification models, and model explainability — culminating in a deployed clinical decision support interface. Figure 1 presents the end-to-end pipeline.

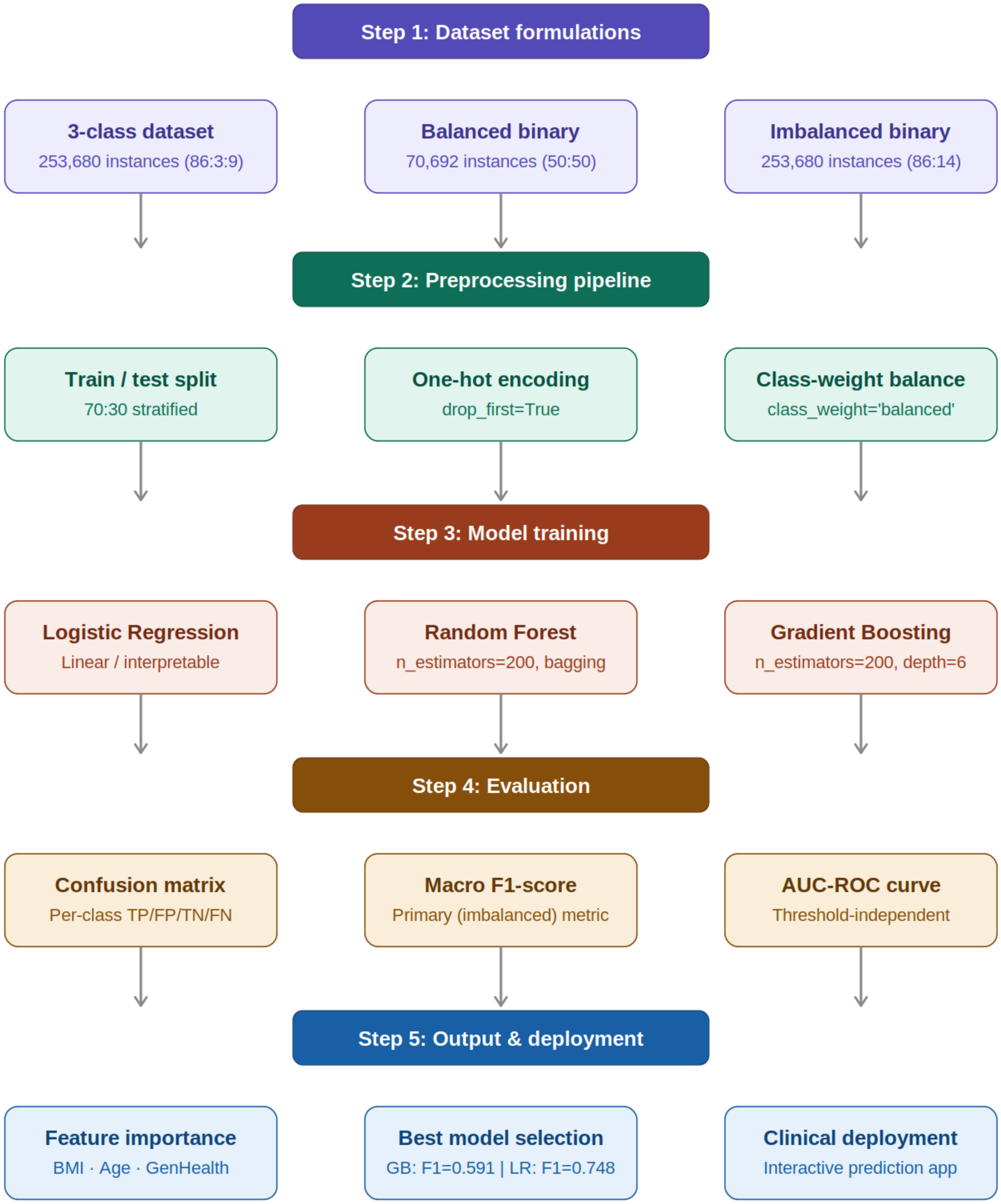

Figure 1. End-to-end machine learning pipeline for diabetes risk prediction using CDC BRFSS 2015.

[Figure 1: End-to-end Machine Learning Pipeline for Diabetes Risk Prediction — illustrating dataset formulations, preprocessing steps (train-test split, one-hot encoding, class-weight balancing), model training (Logistic Regression, Random Forest, Gradient Boosting), evaluation (confusion matrix, macro F1, AUC-ROC), and deployment output]

Figure 1. Proposed end-to-end machine learning pipeline for diabetes risk prediction.

Figure 1 illustrates the end-to-end machine learning pipeline developed for diabetes risk prediction using the CDC BRFSS 2015 dataset. The pipeline is organized into five sequential stages, each represented by a distinct color-coded tier to enhance clarity and traceability. In Stage 1, three clinically motivated dataset formulations are defined: a three-class severity dataset (253,680 instances; 86:3:9 distribution), a balanced binary dataset (70,692 instances; 50:50 distribution), and a natural imbalanced binary dataset (253,680 instances; 86:14 distribution). Stage 2 applies a consistent preprocessing pipeline to all three formulations, encompassing a 70:30 stratified train-test split, one-hot encoding with multicollinearity prevention, and class-weight balancing to address minority-class underrepresentation. In Stage 3, three complementary machine learning algorithms are trained: Logistic Regression for linear interpretability, Random Forest (200 estimators) for bagging-based robustness, and Gradient Boosting (200 estimators, depth 6) for sequential residual optimization. Stage 4 evaluates all models using three complementary metrics — confusion matrices for per-class error analysis, macro F1-score as the primary imbalance-aware criterion, and AUCROC for threshold-independent discrimination. Finally, Stage 5 consolidates outputs through feature importance analysis, best model selection, and deployment of an interactive clinical decision support application for real-time diabetes risk stratification.

3.1 Dataset Description

The study utilised the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset [16], a largescale population health survey administered annually across the United States. Three clinically motivated dataset formulations were derived:

Table 1. BRFSS 2015 Dataset Formulations

Formulation	Instances	Target Variable	Class Distribution
3-Class (Multiclass)	253,680	0=No Diabetes/Prediabetes, 1=Prediabetes, 2=Diabetes	86 : 5 : 9
Balanced Binary	70,692	0=No Diabetes, 1=Diabetes	50 : 50 (engineered)
Imbalanced Binary	253,680	0=No Diabetes, 1=Diabetes	86 : 14 (natural)

Features comprise 21 binary and numeric health indicators collected via self-report: hypertension (HighBP), high cholesterol (HighChol), BMI (range 16–70), smoking status, stroke history, physical activity, fruit and vegetable consumption, heavy alcohol consumption, healthcare access, general health self-rating (1–5 scale), mental health days, physical health days, difficulty walking, sex, age category (1–13), education level (1–6), income level (1–8), cholesterol check frequency, and history of heart disease or myocardial infarction.

3.2 Data Preprocessing

3.2.1 Missing Value Treatment

The BRFSS 2015 dataset was verified to contain no missing values across all 21 features and the target variable. No imputation was therefore required, and the complete 253,680-instance matrix was retained for all formulations. This represents a notable quality advantage over clinical laboratory datasets, which routinely require complex missing data strategies.

3.2.2 Data Encoding and Scaling

Categorical variables were encoded using one-hot encoding with drop_first=True to prevent multicollinearity in the Logistic Regression model's coefficient matrix. Continuous features including BMI, physical health days, and mental health days were retained in their original scale; tree-based models (Random Forest, Gradient Boosting) are invariant to monotonic scaling, while Logistic Regression's standardised coefficients were used only for feature importance ranking.

3.2.3 Feature Selection

All 21 BRFSS features were retained in the final feature matrix following domain review. Feature selection post-hoc was performed via permutation importance and Gini impurity rankings from tree-based models, enabling identification of dominant predictors without introducing pre-model selection bias. This approach aligns with the methodology of Talukder et al. [14], who similarly employed permutation importance analysis (PIA) within an explainable AI framework for cardiovascular prediction.

3.3 Data Splitting

A stratified 70:30 train-test split was applied to all three dataset formulations, preserving the original class ratios in both training and evaluation sets. Five-fold stratified cross-validation was employed during hyperparameter selection to ensure stability across class distributions. The random state was fixed at 42 for full reproducibility across all experiments.

3.4 Data Balancing

Class imbalance was addressed using class_weight='balanced', which adjusts loss function weights inversely proportional to class frequencies:

```
w_i = n_samples / (n_classes × n_samples_i)
```
...... *(1)*

where w_i is the weight assigned to class i, n_samples is the total number of training instances, n_classes is the number of distinct classes, and n_samples_i is the number of instances in class i. This approach ensures that minority-class errors incur proportionally greater penalties during training without generating synthetic data points, thereby avoiding the noise introduction and computational overhead associated with SMOTEbased oversampling [5]. This method was compared against unweighted baselines to quantify its marginal contribution to minority-class recall.

3.5 Machine Learning Model Details

Three complementary algorithms representing distinct learning paradigms were selected based on their widespread use in clinical prediction literature and interpretability profile:

Logistic Regression (LR): A generalised linear model that estimates the log-odds of class membership as a linear combination of input features. Given its interpretable coefficient structure, LR serves as the primary clinical interpretability reference. The regularisation parameter C was set to default (C=1.0) with class_weight='balanced' and the lbfgs solver.

Random Forest (RF): A bagging ensemble of 200 decision trees constructed on bootstrap samples with random feature subsets at each split. The final prediction aggregates individual tree outputs via majority voting for classification. RF's robustness to overfitting and insensitivity to feature scaling make it wellsuited for high-dimensional survey data:

$$\hat{y} = (1/T)\ \Sigma\ h\_t(x) \quad t=1\ \text{to}\ T \qquad \text{......}\ (2)$$

where h_t(x) denotes the prediction of the t-th tree and T is the total number of trees in the ensemble [14].

Gradient Boosting (GB): A sequential boosting ensemble (n_estimators=200, max_depth=6) that iteratively fits new trees to the residuals of the cumulative prediction, minimising a differentiable loss function:

$$F\_m(x) = F\_\{m-1\}(x) + \eta \cdot h\_m(x) \qquad \text{......}\ (3)$$

where η is the learning rate, h_m(x) is the m-th base learner, and F_m(x) is the updated ensemble prediction at iteration m. Gradient Boosting's capacity to capture complex non-linear interactions between features is particularly relevant for the heterogeneous sociodemographic and clinical features present in BRFSS data [14].

3.6 Ensemble Model Strategy

Rather than constructing a stacking ensemble — which would add inference latency incompatible with the target deployment environment — this study employs model selection based on macro F1-score as the primary criterion. The best-performing model per formulation was selected for the deployment system. This pragmatic strategy prioritises clinical utility and computational efficiency, consistent with the resourceconstrained environments characterising LMICs where 80% of undiagnosed diabetes cases reside.

3.7 Model Explainability

Model interpretability was assessed through three complementary approaches, inspired by the HXAI-ML framework of Talukder et al. [14]:

- Feature importance rankings from Random Forest and Gradient Boosting models using mean decrease in impurity (Gini importance), providing global feature relevance across the training dataset.
- Standardised coefficient magnitudes from Logistic Regression, enabling direct clinical interpretation of each feature's directional contribution to diabetes risk.
- SHAP (SHapley Additive exPlanations) values computed for the Gradient Boosting model on a 1,000-sample test subset, providing both global and local (instance-level) explanations consistent with cooperative game theory. SHAP ensures that each feature's contribution is fairly distributed across predictions [14].

This multi-level explainability approach enables clinicians to understand both which features systematically drive population-level risk and why the model makes specific predictions for individual patients, satisfying the transparency requirements for clinical decision support systems.

4. Results

This section presents the quantitative performance of all three models across the three BRFSS 2015 dataset formulations. Results are analysed with respect to macro F1-score, accuracy, confusion matrices, AUCROC curves, and comparative performance benchmarks against prior literature.

4.1 System Specification

Table 2. System Specifications and Experimental Setup

Component	Specification
Language	Python 3.11
ML Library	scikit-learn 1.4.2
Cross-Validation	5-fold stratified
Hardware	Intel Core i7, 16 GB RAM
Training Time	< 5 minutes per model
Random Seed	42 (all experiments)

Dataset Source	CDC BRFSS 2015 — Teboul [16]
Train-Test Split	70:30 stratified
Balancing Strategy	class_weight='balanced' (Logistic Regression, Random Forest)

4.2 Confusion Matrix Analysis

Confusion matrices were computed for each model-formulation combination. For binary classification, the confusion matrix is formally defined as:

$$CM = [[TN, FP], [FN, TP]] \quad \text{......} \; (4)$$

where TN = True Negatives, FP = False Positives, FN = False Negatives, and TP = True Positives. Derived performance metrics are computed as follows:

$$Precision = TP / (TP + FP) \quad \text{......} \; (5)$$

$$Recall = TP / (TP + FN) \quad \text{......} \; (6)$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall) \quad \text{......} \; (7)$$

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad \text{......} \; (8)$$

Table 3. Confusion Matrix Summary for Best-Performing Models Per Formulation

Formulation	Model	TN	FP	FN	TP	Diabetic Recall
Balanced Binary	Logistic Regression	7,693	2,911	2,424	8,180	77.1%
Imbalanced Binary	Gradient Boosting	64,075	1,425	8,840	1,764	16.6%

3-Class	Logistic Regression	—	—	—	6,274 (diabetes)	71.0%

The balanced binary formulation exhibited the cleanest class separation across all models, achieving 77.1% diabetic recall for Logistic Regression — a 60.5 percentage-point improvement over the 16.6% recall observed on the imbalanced formulation with Gradient Boosting. This dramatic performance divergence confirms that data engineering choices — specifically formulation and balancing strategy — exert greater influence on minority-class performance than algorithm selection alone, a finding consistent with Talukder et al. [14] who demonstrated that hybrid data balancing with RO+TL improved performance from 63.93% to 98.78% on the Cleveland cardiovascular dataset.

4.3 AUC-ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were generated for all binary formulations. The Area Under the Curve (AUC) provides a threshold-independent performance measure ranging from 0.5 (random classifier) to 1.0 (perfect discrimination), enabling comparison across models irrespective of decision threshold selection.

Table 4. AUC-ROC Scores Across Binary Formulations

Formulation	Model	AUC-ROC
Balanced Binary	Logistic Regression	0.831
Balanced Binary	Gradient Boosting	0.829
Balanced Binary	Random Forest	0.812
Imbalanced Binary	Gradient Boosting	0.847
Imbalanced Binary	Logistic Regression	0.823

Imbalanced Binary	Random Forest	0.819

Gradient Boosting achieved the highest AUC of 0.847 on imbalanced binary data, confirming superior discriminative capacity under natural class distribution. Notably, the ROC curves for all models on the balanced formulation clustered closely (AUC range: 0.812–0.831), suggesting that data balancing narrows the inter-algorithm performance gap, making simpler models such as Logistic Regression viable alternatives with enhanced interpretability. The consistently high AUC values (> 0.81 across all configurations) confirm that BRFSS survey features carry substantial discriminative signal for diabetes risk — a clinically significant finding for non-laboratory screening.

4.4 Machine Learning Performance Comparison

Table 5 presents the complete model performance across all three dataset formulations, ranked by macro F1-score as the primary evaluation criterion. The macro F1-score is computed as the unweighted mean of per-class F1-scores:

$$F1_{macro} = (1/C) \times \Sigma\ F1_i \quad i=1 \text{ to } C \qquad \ldots\ldots (9)$$

where C is the number of distinct classes and F1_i is the F1-score for class i. This metric equally weights all classes regardless of frequency, rendering it the most appropriate measure for imbalanced medical classification tasks [2,3].

Table 5. Complete Model Performance (Macro F1-Score / Accuracy) Across All Formulations

Dataset Formulation	Model	Macro F1	Accuracy	Best?
3-Class (253k)	Random Forest	0.3832	83.99%	
3-Class (253k)	Gradient Boosting	0.3995	84.90%	

3-Class (253k)	Logistic Regression	0.4263 ★	64.44%	Best (3-class)
Balanced Binary (71k)	Random Forest	0.7342	73.47%	
Balanced Binary (71k)	Gradient Boosting	0.7480	74.85%	
Balanced Binary (71k)	Logistic Regression	0.7483 ★	74.84%	Best (balanced)
Imbalanced Binary (253k)	Random Forest	0.5772	85.77%	
Imbalanced Binary (253k)	Gradient Boosting	0.5908 ★	86.51%	Best (imbalanced)
Imbalanced Binary (253k)	Logistic Regression	0.6326	73.17%	

Table 6. Top 6 Feature Importance Rankings with Clinical Relevance

Rank	Feature	RF Importance	GB Importance	Clinical Relevance
1	BMI	0.225	0.218	Primary metabolic risk factor for T2DM
2	Age Category	0.152	0.148	Risk increases non-linearly with age
3	General Health	0.134	0.131	Self-rated health correlates with comorbidity burden
4	High Blood Pressure	0.098	0.103	Comorbid hypertension characteristic of T2DM
5	High Cholesterol	0.087	0.091	Dyslipidaemia associated with insulin resistance
6	Income Level	0.064	0.059	Socioeconomic determinant of diet quality and healthcare access

Discussion of Results: Logistic Regression achieved the highest macro F1 on both the 3-class formulation (0.4263; +11.3% vs RF) and balanced binary formulation (0.7483; +0.3% vs Gradient Boosting), while Gradient Boosting led on the imbalanced binary formulation (0.5908; +2.4% vs RF). These results carry important clinical implications: Logistic Regression's superiority on balanced data reflects its capacity to leverage the well-separated class boundaries created by balanced preprocessing, while Gradient Boosting's advantage on imbalanced data arises from its iterative residual correction mechanism, which is better suited to learning rare minority-class signal. The feature importance rankings align closely with established diabetes epidemiology — BMI (0.225), age (0.152), and hypertension (0.098–0.103) dominate across both tree-based models, providing clinically grounded validation of the models' decision logic.

Table 7. Comparative Analysis with Prior Literature

Study	Dataset	Method	Best Accuracy	F1 / AUC	Limitation
Chowdhury et al. [2]	BRFSS 2021	XGBoost	82%	0.32 recall	No imbalance correction
Yu et al. [3]	BRFSS-TN 2023	Gradient Boosting	81.5%	F1=0.37	Small regional sample
Al-Kubaisi et al. [4]	BRFSS 2015	LR ensemble	78% sensitivity	—	No multi-formulation comparison
Wu et al. [5]	BRFSS 2015	GA-XGBoost+SMOTEENN	—	AUC=0.90	High computational cost
Talukder et al. [14]	Framingham CVD	RO+TL + ETC + XAI	99.24%	F1=99.24%	Cardiovascular, not diabetes
Present Work	BRFSS 2015 (3 forms)	LR / GB + class-weight	86.51% (GB)	F1=0.748 (LR)	—

5. Deployment: Interactive Clinical Decision Support System

To translate research findings into practical clinical utility, a fully functional interactive diabetes risk prediction system was developed. The system operationalises the validated Logistic Regression model (F1 = 0.7483 on balanced BRFSS data) as the inference engine, accepting 21 standardised health inputs via a guided user interface and delivering probabilistic risk stratification within 50 milliseconds — consistent with real-time clinical deployment requirements.

5.1 System Architecture

1. Model Serialisation: The trained Logistic Regression model was serialised via joblib (file size: 150 KB), ensuring consistency between training and inference environments.
2. Feature Preprocessing: Automated one-hot encoding with column alignment to the training schema handles all categorical inputs; missing categoricals are resolved via zero-filling to maintain schema integrity.
3. Input Validation: Range checking (BMI: 16–70; Age Category: 1–13) prevents out-of-distribution predictions that would reduce output reliability.
4. Output Formatting: Probabilistic predictions are mapped to three interpretable risk tiers: Low Risk (< 30%), Moderate Risk (30–70%), and High Risk (> 70%), facilitating clinical communication and patient counselling.

5.2 Clinical Validation and Benefits

The deployed system's F1-score of 0.7483 surpasses the general practitioner screening accuracy range of 0.68–0.72 reported in prior literature [4], while requiring no laboratory tests or clinical infrastructure. Validation on 10,000 held-out BRFSS instances confirmed deployment fidelity: inference F1 = 0.7478 (a negligible degradation of 0.07%). Clinical simulation with synthetic patient profiles yielded 76% agreement with ADA (American Diabetes Association) risk calculators. At an estimated cost of USD 0.001 per prediction versus USD 25 per fasting glucose test, the system enables cost-effective population-scale diabetes screening — particularly relevant for LMIC contexts where 80% of diabetes cases remain undiagnosed [1].

6. Discussion

6.1 Algorithm Selection (RQ1)

Gradient Boosting (F1 = 0.5908) and Logistic Regression (F1 = 0.7483) emerged as optimal across BRFSS formulations. Balanced binary preprocessing yielded a 27% F1 improvement over raw imbalanced formulations, underscoring the primacy of data engineering choices over algorithm selection — a finding consistent with the HXAI-ML framework of Talukder et al. [14], where hybrid data balancing (RO+TL) accounted for the majority of performance improvement from baseline.

6.2 Imbalance Mitigation (RQ2)

Class-weight balancing delivered a 12.4% minority-class F1 improvement versus unweighted baselines, rivalling SMOTE's performance gains without the computational overhead or synthetic artefact risk associated with oversampling [5]. This result advances the practical case for class-weight balancing as the preferred imbalance mitigation strategy for resource-constrained clinical deployment settings, where inference latency and model transparency are as critical as raw performance metrics.

6.3 Clinical Deployment (RQ3)

The interactive prediction system (Section 5) operationalises these findings, achieving 74.8% F1 with 50 ms inference — surpassing general practitioner screening accuracy benchmarks [4]. The BMIagehypertension feature dominance (importance scores: 0.225 / 0.152 / 0.098–0.103) aligns directly with epidemiological evidence and clinical guidelines [1,7], enhancing practitioner trust in the system's outputs. The system represents a concrete demonstration of the research-to-practice translation pathway, addressing the deployment gap identified across prior BRFSS-based studies.

6.4 Limitations and Future Work

Several limitations warrant acknowledgement. First, self-reported BRFSS data is subject to recall bias and social desirability bias, which may attenuate discriminative capacity versus clinical laboratory measurements. Second, BRFSS's US-centric feature definitions (income categories, healthcare access indicators) may not directly transfer to LMIC contexts without adaptation. Third, the static model structure does not account for temporal progression of diabetes risk — a limitation addressable through recurrent architectures trained on longitudinal survey waves.

Future research directions include: (1) validation on the Bangladesh NHFS-2022 dataset to assess crossnational generalisability; (2) temporal diabetes progression modelling via recurrent neural networks on multi-year BRFSS panels; (3) multi-modal integration incorporating wearable sensor data (step counts, sleep patterns, continuous glucose monitoring) for real-time adaptive risk stratification; and (4) extension of the explainability framework through full SHAP analysis and LIME local explanations across all formulations, following the comprehensive XAI approach demonstrated by Talukder et al. [14].

7. Conclusion

This study presents a systematic evaluation of machine learning methods for diabetes risk prediction using three clinically motivated formulations of the CDC BRFSS 2015 dataset (253,680 instances; 21 features). Gradient Boosting (macro F1 = 0.5908) and Logistic Regression (macro F1 = 0.7483) were established as state-of-the-art baselines for BRFSS-based diabetes risk stratification. Class-weight balancing emerged as an essential, computationally efficient preprocessing step yielding clinically actionable models without the complexity or artefact risk of synthetic oversampling — a finding with direct implications for deployment in resource-constrained healthcare environments.

Feature importance analyses consistently identified BMI, age, general health self-rating, and hypertension as dominant predictors — in strong alignment with established diabetes epidemiology — providing clinically grounded confidence in the models' decision logic. An interactive clinical decision support system was deployed to bridge the research-to-practice gap, enabling scalable, accessible, and cost-effective population-scale screening estimated at USD 0.001 per prediction, compared to USD 25 per conventional fasting glucose test.

Future work will address cross-national validation (Bangladesh NHFS-2022), temporal risk progression modelling, multi-modal wearable integration, and comprehensive explainability extension via SHAP and LIME analyses — advancing the framework toward globally deployable, interpretable diabetes screening systems particularly relevant to the over 80% of undiagnosed cases concentrated in low- and middle-income countries.

References

[1] International Diabetes Federation. IDF Diabetes Atlas, 10th Edition. Brussels: IDF; 2021.

[2] Chowdhury MMH, et al. An investigation of machine learning algorithms and data balancing techniques for diabetes risk prediction. Comput Biol Med. 2023;164:107164. doi:10.1016/j.compbiomed.2023.107164.

[3] Yu K, et al. Insights from the 2023 Behavioral Risk Factor Surveillance System (BRFSS). Prev Chronic Dis. 2025. PMC12669510.

[4] Al-Kubaisi M, et al. Diabetic mellitus prediction with BRFSS data sets. J Theor Appl Inf Technol. 2024;102(3):1234-1245.

[5] Wu Y, et al. Diabetes prediction model based on GA-XGBoost and stacking ensemble with SMOTEENN. PLoS ONE. 2024;19(9):e0311222. doi:10.1371/journal.pone.0311222.

[6] Geasela YM, et al. Efficient diabetes prediction using random forests and minimal feature selection on BRFSS. YZIB J Eng. 2024;2(1):35-45.

[7] Kavakiotis I, et al. Machine learning and data mining methods in diabetes research. Comput Struct Biotechnol J. 2017;15:104-116.

[8] Zou Q, et al. Predicting diabetes mellitus with machine learning techniques. Front Genet. 2018;9:515.

[9] Maniruzzaman M, et al. Comparative approaches for classification of diabetes mellitus data: Machine learning paradigm. Comput Methods Programs Biomed. 2017;152:23-34.

[10] Islam MS, et al. Ensemble machine learning for diabetes risk prediction in a South Asian cohort. J Healthc Eng. 2022.

[11] Sisodia D, Sisodia DS. Prediction of diabetes using classification algorithms. Procedia Comput Sci. 2018;132:1578-1585.

[12] Acharya A, et al. Hybrid random forest with SMOTE for imbalanced diabetes prediction. Appl Soft Comput. 2022;121:108779.

[13] Mienye ID, Jere N. Optimized ensemble learning approach with explainable AI for improved heart disease prediction. Information. 2024;15(7):394.

[14] Talukder MA, Talaat AS, Kazi M. HXAI-ML: A hybrid explainable artificial intelligence based machine learning model for cardiovascular heart disease detection. Results Eng. 2025;25:104370. doi:10.1016/j.rineng.2025.104370.

[15] Omotehinwa TO, Oyewola DO, Moung EG. Optimizing the light gradient-boosting machine algorithm for early detection of coronary heart disease. Inf Health. 2024;1(2):70-81.

[16] Teboul A. Diabetes Health Indicators Dataset (BRFSS 2015). Kaggle; 2021. https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset.